LAB 05: INTRODUCTION TO FOR LOOPS AND SWITCH CASE

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Task 1: C++ program to print the table of any number

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= 10; i++)
        cout << num << " x " << i << " = " << num * i << endl;
    return 0;
}
```

Task 2: Find the power of any number using a for loop

```
#include <iostream>
using namespace std;
int main() {
    int base, power, result = 1;
    cout << "Enter base and power: ";
    cin >> base >> power;
    for (int i = 1; i <= power; i++)
        result = result * base;
    cout << "Answer = " << result << endl;
    return 0;
}
```

Task 3: Sum of natural numbers using a loop

```
#include <iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

Task 4: Find the sum of digits of a number

```
#include <iostream>
using namespace std;
int main() {
    int num, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    while (num > 0) {
        sum = sum + num % 10;
        num = num / 10;
    }
    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

Task 5: Generate the factorial of a number

```
#include <iostream>
```

```
using namespace std;
int main() {
    int n, fact = 1;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        fact = fact * i;
    cout << "Factorial = " << fact << endl;
    return 0;
}
```

Task 7: Print the given star pattern

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }
    return 0;
}
```

Output

```
*
**
***
****
*****
******
*******
```

Task 8: Print the total number of days in a month using switch case

```
#include <iostream>
using namespace std;
int main() {
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;
    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << "31 days"; break;
        case 4: case 6: case 9: case 11:
            cout << "30 days"; break;
        case 2:
            cout << "28 days (29 in a leap year)"; break;
        default:
            cout << "Invalid month";
    }
    return 0;
}
```

Task 9: Check whether an alphabet is a vowel or consonant using switch case

```
#include <iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << "Vowel"; break;
        default:
            if ((ch >= 'a' && ch <= 'z') ||
                (ch >= 'A' && ch <= 'Z'))
                cout << "Consonant";
            else
                cout << "Not an alphabet";
    }
    return 0;
}
```

Think Questions: Outputs

1. Given code: for (5; 2; 2) { cout << "Hello"; }

Answer: It prints Hello continuously (an infinite loop), because the condition expression 2 is always true. The snippet also needs the appropriate C++ header for cout.

2. Given code: for (i = 0; i > 9; i += 3) { cout << "for "; } (input is 6)

Answer: No output. The loop starts with i = 0, and 0 > 9 is false. The stated input of 6 is not used by this code.

3. Given code: for (;;) { cout << "10"; }

Answer: It prints 10 continuously without stopping because all three for-loop expressions are omitted, making it an infinite loop.

4. Given code: int fun=5; cout << "C++ for loop "; for (x=0; x<=fun; x++) cout << x;

Output:

```
C++ for loop 012345
```

Note: The worksheet labels the star pattern as Task 7 and then lists two additional switch-case tasks. Both switch-case tasks are included above.